

Applied Optics for Fiber Optic Communications: Theory and Practice – Part 2

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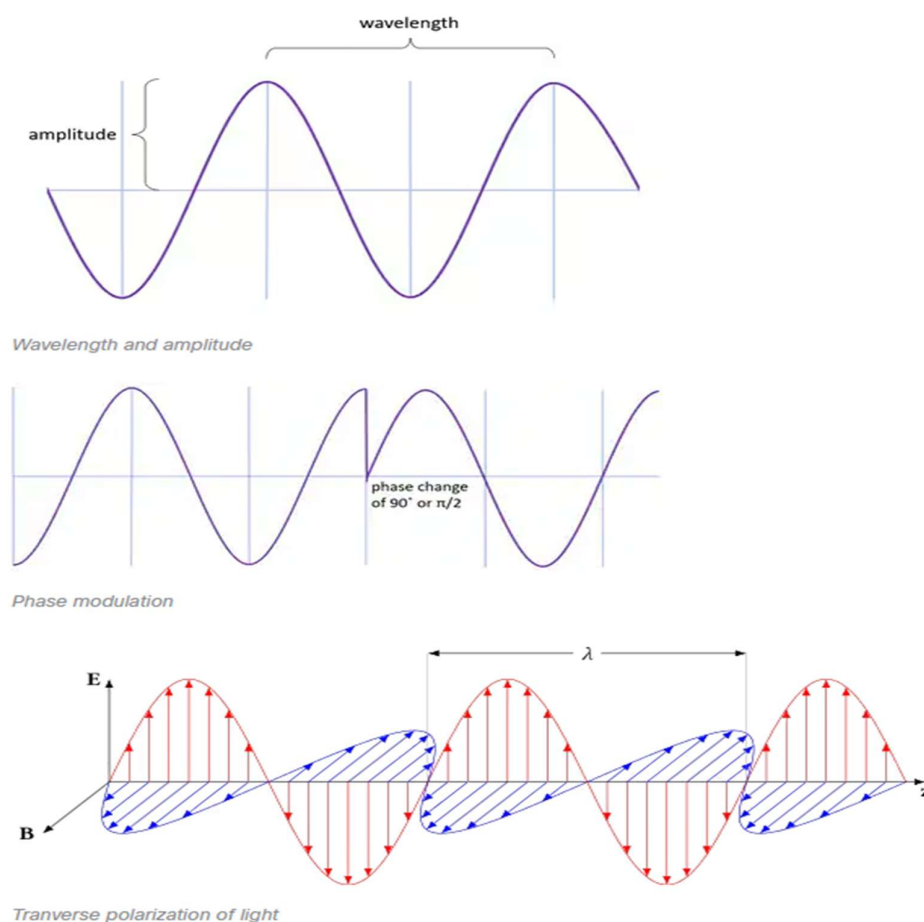
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Chapter 3: Foundational Optical Topics

A modulation scheme continuously alters the property or properties of a waveform. In our case, it is light, to encode the binary information into the waveform. Modern optical networks use a variety of modulation schemes to transport the data across hundreds to thousands of kilometres. Fiber optical networks utilize several properties of light to efficiently encode the information

1. **Wavelength** or frequency - each channel in a fibre optical network uses a specific wavelength in the C/L-band, between approximately 1527 nm and 1565 nm. Each signal can provide varying bandwidth depending on the baud rate and modulation scheme.
2. **Phase** - the angle of a waveform typically measured in radians. Changing the phase translates the period of the waveform in time.
3. **Amplitude** - a measure of the total power of a signal in decibel-milliwatts (dBm).
4. **Polarization** - electromagnetic waves have two primary polarization states defined by the electric and magnetic fields. Each polarization can contain information encoded by a modulation scheme. Some Cisco optical products use the notation Coherent Polarization-Multiplexed (CP) or Polarization Multiplexed (PM) in order to identify the use of polarization in the modulation.



3.1 Modulation

Modulation is the **Fundamental Process** in Communication Systems, In Modulation **Low-Frequency Message Signal Blends** with **High-Frequency Carrier Frequency** to **Enable Efficient Transfer** of the Signal

Modulation is the process of **superimposing high-frequency carrier signals** with **low-frequency message signals** resulting in a **modulated wave**. In modulation, **one of the three parameters** of the **carrier wave is changed** i.e. **amplitude, frequency or phase** according to the amplitude of the message signal at a given time instant. It is an important technique used to increase the range of communication, increase the signal-to-noise ratio, and decrease the size of the antenna.

There are mainly two types of modulation techniques which are

1. Analog Modulation
2. Digital Modulation

3.1.1 Analog Modulation

In Analog Modulation, both **carrier and message signals are analogue waves**. Analog waves are those signals which can take an infinite value between two points



3.1.2 Digital Modulation

In Digital Modulation, only the carrier signal is analogue, and the message signal is in digital form. Digital signals are binary values that can take either logic 0 as in zero volts or logic 1 as in any other voltage. There are no in-between voltage levels in the digital signal

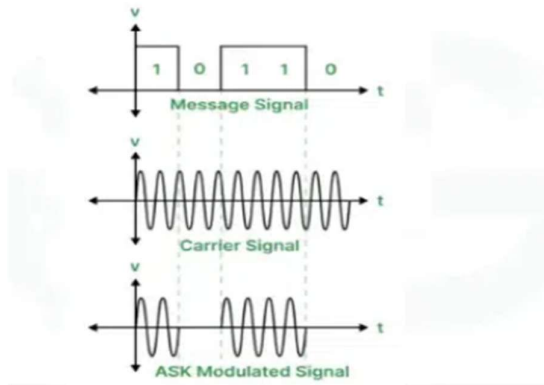
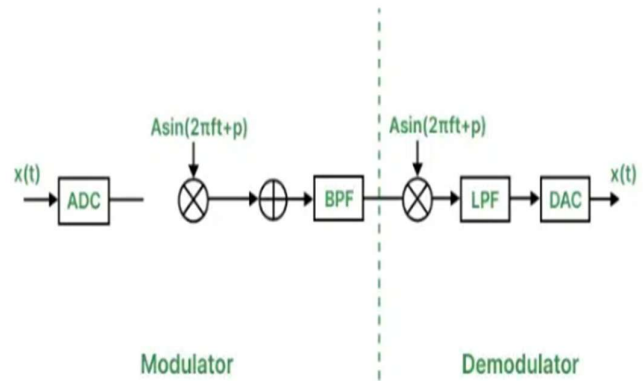


Shift Keying means that the amplitude, frequency or phase of the carrier wave is shifted between two or more discrete values rather than varying continuously like Analog Modulation. Binary data requires two discrete levels of amplitude, frequency or phase for modulation called as Binary Shift Keying. A group of bits can be clubbed together to form M-ary Shift Keying. There are mainly three types of Digital Modulation techniques. They are

3.1.2.1 Amplitude Shift Keying

1. **Amplitude Shift Keying:** each symbol in the message signal gives a unique amplitude to the carrier wave. There are two types of ASK, Binary and M-ary.
 - In Binary ASK, logic 1 is associated with certain amplitude of carrier wave e.g. 12V and logic 0 is associated with different amplitude other than 12V e.g. 0V.
 - In M-ary ASK, a group of $\log_2 M$ bits are considered together rather than 1 bit at a time and the amplitude level is associated with this group of bits

For example, in 16-ary ASK, a group of 4 bits are considered and are given a respective amplitude. Since there are 16 possible 4-bit binary numbers (2^4), 16 different amplitude levels are required for modulation. If all such amplitudes are created using a single carrier wave, then it is called as coherent ASK. If multiple carrier wave each with different amplitudes are used for modulation, then it is called as non-coherent ASK.

Amplitude Shift Keying Waveform**Block Diagram of Amplitude Shift Keying****Modulation Process****1. Analog-to-Digital Conversion:**

The analog message signal is digitized using an **Analog-to-Digital Converter (ADC)**.

2. Carrier Multiplication:

- The digital signal **$m(t)$** is fed to a multiplier circuit.
- A high-frequency sinusoidal carrier wave **$A \sin(2\pi f_c t + \phi)$** is multiplied with **$m(t)$** , producing: **$s(t) = m(t) \cdot A \sin(2\pi f_c t + \phi)$**
- Logic States:
 - **High (1):** Carrier wave passes unchanged.
 - **Low (0):** Output becomes zero (no signal).
- Bandwidth Optimization:
 - The abrupt amplitude changes in the ASK signal create high-frequency components, leading to excessive bandwidth usage.
 - A Bandpass Filter (BPF) smoothens the signal, limiting bandwidth while preserving the encoded data.

Demodulation Process**1. Carrier Re-Multiplication:**

The received ASK signal is multiplied again by the same carrier wave, resulting in:

$$s_{\text{demod}}(t) = m(t) \cdot A \sin^2(2\pi f_c t + \phi)$$

2. Signal Recovery:

A Low-Pass Filter (LPF) removes high-frequency components, leaving the baseband signal **$m(t)$** .

3. Digital-to-Analog Conversion:

The filtered digital signal is converted back to analog using a **Digital-to-Analog Converter (DAC)**.

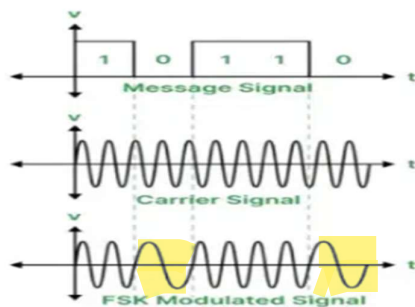
3.1.1.2 Frequency Shift Keying

In Frequency Shift Keying (FSK), each symbol in the message signal gives a unique frequency to the carrier wave. There are two types of FSK, Binary and M-ary.

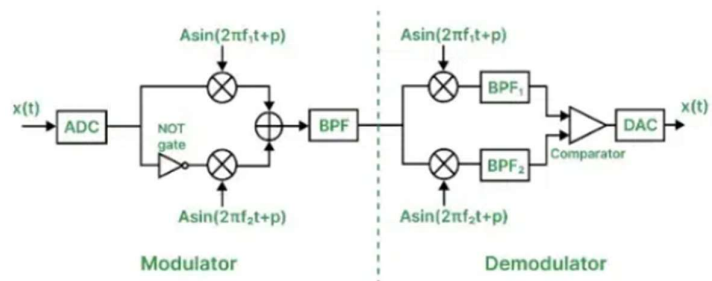
- In Binary FSK, logic 1 is associated with certain frequency of carrier wave e.g. 50MHz and logic 0 is associated with different frequency other than 50MHz e.g. 25MHz.
- In M-ary FSK, a group of $\log_2 M$ bits are considered together rather than 1 bit at a time and the frequency is associated with this group of bits.

For example, in 16-ary FSK, a group of 4 bits are considered and are given a respective frequency. Since there are 16 possible 4 bit binary numbers (24), 16 different frequencies are required for modulation. If all such frequencies are created using a single carrier wave, then it is called as **coherent FSK**. If multiple carrier waves each with different frequencies are used for modulation, then it is called **non-coherent FSK**.

Frequency Shift Keying Waveform



Block Diagram of Frequency Shift Keying



Modulation Process

1. Analog-to-Digital Conversion:

The analog message signal is digitized using an Analog-to-Digital Converter (ADC).

2. Dual-Carrier Multiplication:

- The digital signal splits into two parallel paths:
 - **Logic 1:** Multiplied by Carrier 1: $Asin(2\pi f_1 t + \phi) Asin(2\pi f_1 t + \phi)$.
 - **Logic 0:** Inverted via a **NOT gate** and multiplied by Carrier 2: $Asin(2\pi f_2 t + \phi) Asin(2\pi f_2 t + \phi)$.
- **Result:**
 - For **logic 1:** Only $A \sin(2\pi f_1 t + \phi) + \sin(2\pi f_1 t + \phi)$ is generated (another path gets 0 input).
 - For **logic 0:** Only $A \sin(2\pi f_2 t + \phi) + \sin(2\pi f_2 t + \phi)$ is generated (original path gets 0 input).

3. Signal Combination:

The outputs of both multipliers are summed to create the FSK waveform:

$$s(t) = A[\sin(2\pi f_1 t + \phi) + \sin(2\pi f_2 t + \phi)]s(t) + [\sin(2\pi f_1 t + \phi) + \sin(2\pi f_2 t + \phi)]$$

4. Bandwidth Optimization:

- Sudden frequency shifts in the FSK signal create unwanted high-frequency components.
- A Bandpass Filter (**BPF**) smoothens the waveform, reducing bandwidth usage.

3.1.2.3 Phase Shift Keying

Phase Shift Keying (PSK) is a digital modulation technique where the phase of a carrier wave is altered to encode data. It enables efficient data transmission by mapping digital bits to distinct phase states of the signal, making it fundamental in modern communication systems like optical fibers, wireless networks, and satellite communications

PSK transmits data by varying the phase of a sinusoidal carrier wave. Each phase shift corresponds to a unique combination of binary digits for example

- **Binary PSK (BPSK)** uses two phases (0° and 180°) to represent 0 and 1
- **Quadrature PSK (QPSK)** uses four phases (0° , 90° , 180° , 270°) to encode two bits per symbol

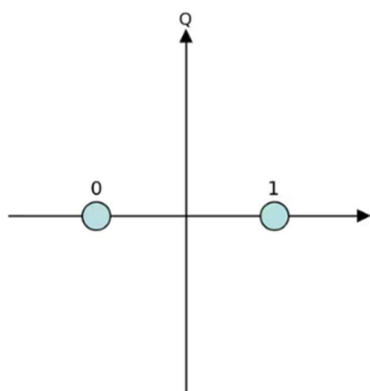
This phase modulation improves bandwidth efficiency and noise resilience compared to analog methods

3.1.2.3.1 Binary Phase Shift Keying (BPSK)

- **Modulation:** The carrier wave shifts phase by 180° for binary inputs (e.g., 0° for 0, 180° for 1). Mathematically, the modulated signals are

$$s_0(t) = A \cos(2\pi f_c t) \quad (\text{for bit 0}).$$

$$s_1(t) = -A \cos(2\pi f_c t) \quad (\text{for bit 1}).$$



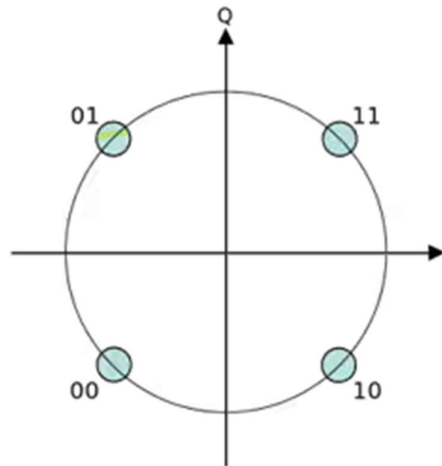
Constellation diagram of BPSK modulation

- **Demodulation:** The receiver compares the phase of the incoming signal with a reference carrier to decode bit
- **Applications**
 - **Satellite Communication:** Robust performance in noisy environments.
 - **RFID and Bluetooth:** Simple implementation for short-range data transfer

3.1.2.3.2 Quadrature Phase Shift Keying (QPSK)

- **Modulation:** Splits data into two parallel streams (even and odd bits). Each stream modulates a carrier wave (in-phase and quadrature-phase components) shifted by 90° . The combined output transmits two bits per symbol

$$s_m(t) = A \cos \left(2\pi f_c t + \frac{\pi(2m+1)}{4} \right), \quad m = 0, 1, 2, 3.$$



Constellation diagram of QPSK modulation

- **Demodulation:** Uses two correlators to separate in-phase and quadrature components, recovering the original bits

Feature	BPSK	QPSK
Data Rate	1 bit/symbol	2 bits/symbol
Bandwidth	Higher requirement	50% less than BPSK
Noise Immunity	Better	Slightly weaker

- **Applications**

- **Optical Fiber Communication:** *High-speed data transmission over long distances.*
- **5G and Wi-Fi 6:** Balances speed and reliability in dense networks

3.1.2.4 Key Takeaways

- **PSK Variations:**
 - **BPSK:** Simpler, robust in noise, but slower.
 - **QPSK:** Faster, bandwidth-efficient, but more complex
- **Trade-offs:** Higher-order PSK (e.g., 8-PSK) increases data rates but reduces noise tolerance
- **Modern Use Cases:**
 - **BPSK:** GPS, deep-space communication.
 - **QPSK:** Streaming services, military radios, and fibre-optic backbones

By encoding data in phase shifts, PSK enables reliable digital communication across diverse mediums, from optical fibres to satellite links. Its adaptability ensures relevance in evolving technologies like 5G and IoT

3.2 Quadrature Amplitude Modulation (QAM)

QAM is a technique that combines amplitude and phase modulation to transmit digital data efficiently. It encodes information by varying both the amplitude (strength) and phase (timing) of a carrier wave. This allows more data to be packed into each signal, making it ideal for high-speed communication systems like fibre optics.

QAM can achieve a higher rate than common amplitude modulation and phase modulation, which support only two types of symbols to distinguish 0 and 1.

- **Amplitude modulation:** distinguishes 0 and 1 by changing the carrier amplitude.
- **Phase modulation:** distinguishes 0 and 1 by changing the carrier phase. For example, binary phase shift keying (BPSK) uses phases 0° and 180° for transmitting a single bit (0 or 1) of information. In contrast, QPSK can encode 2 bits per symbol (00, 01, 10, or 11) through the following four phases: 0° , 90° , 180° , and 270° . In fact, QPSK is a special type of QAM, that is, 4-QAM.

QAM can modulate more symbols, and each symbol has its own phase and amplitude.

3.2.1 How QAM Works

1. Carrier Waves:

QAM uses two orthogonal carrier waves (sine and cosine) that are 90° out of phase with each other:

- **In-phase (I):** Modulated by one data stream.
- **Quadrature (Q):** Modulated by another data stream.

2. Modulation:

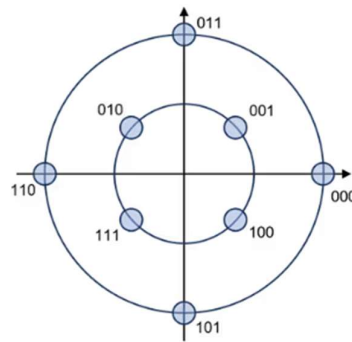
- The I and Q signals are combined into a single waveform.
- Each unique combination of amplitude and phase represents a symbol, which encodes multiple bits of data.

3. Constellation Diagram:

- Symbols are plotted on a grid (e.g., 16-QAM has 16 points, representing 4 bits/symbol).
- Higher-order QAM (e.g., 256-QAM) encodes more bits per symbol but requires better signal quality.

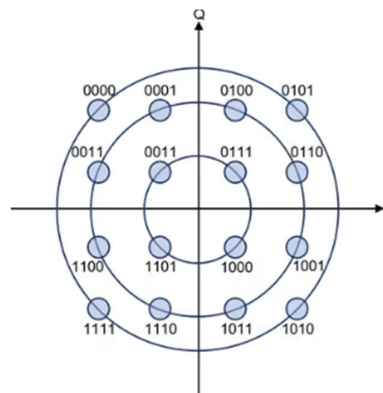
Example:

- **8-QAM:** Eight possible states give **three bits per symbol** for this modulation scheme.



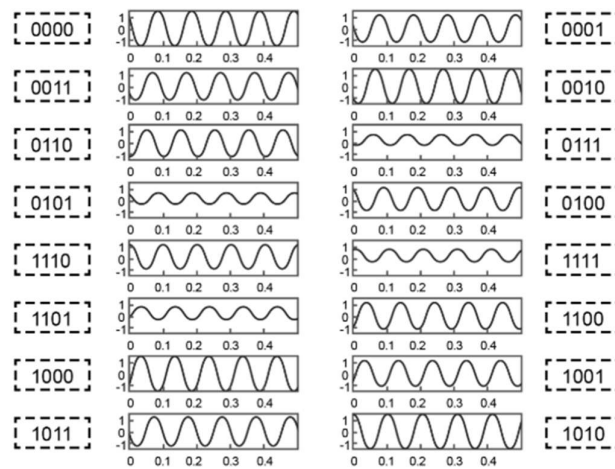
Constellation diagram of 8-QAM

- **16-QAM:** 4 bits/symbol $\rightarrow$ 16 possible symbols. At baud rates around 30 Gbaud, 16-QAM has a data rate of **200 Gbps**. Increasing to **60 Gbaud** gives rates up to **400 Gbps**. Smaller changes in phase and amplitude increase the **OSNR requirements** and limit its **range to a few hundred kilometers**

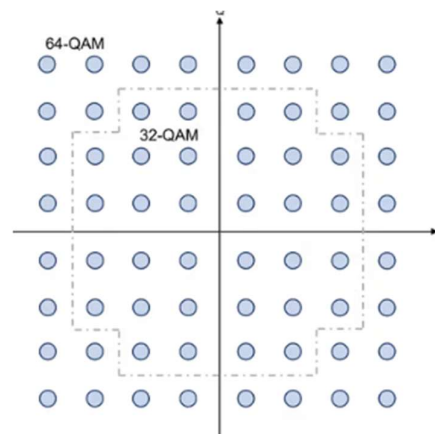


Constellation diagram of 16-QAM

- E.g. **16-QAM** can modulate symbols into 16 different waveforms, representing 0000, 0001, and so on, as shown in the following figure. This means that a total of 16 types of symbols is available, each of which may transmit **4-bit information**.



- **32-QAM and 64-QAM:** These two high-order modulation schemes use **five and six bits** per symbol, respectively, enabling transmission rates up to **600 Gbps**. The high OSNR requirements of **64-QAM** limit the effective range to less than **200 km**.



Constellation diagram of 32-QAM and 64-QAM

3.2.2 QAM in Fiber Optic Communications

QAM is widely used in **coherent optical systems** for high-speed data transmission over fibre. Here's why:

1. **High Spectral Efficiency:**
 - QAM increases data rates by packing more bits per symbol. For example:
 - **16-QAM:** 4 bits/symbol → 100 Gbps in fibre optics.
 - **256-QAM:** 8 bits/symbol → Higher capacity for metro/transport networks.
2. **Coherent Detection:**
 - Modern fibre systems use digital signal processing (DSP) to **decode QAM signals**, compensating for distortions like **phase noise and chromatic dispersion**.
3. **DWDM Compatibility:**
 - QAM works with **Dense Wavelength Division Multiplexing (DWDM)** to maximize fibre bandwidth. Higher-order QAM (e.g., **1024-QAM**) supports **terabit-scale data rates**.

3.2.3 Advantages

- **Bandwidth Efficiency:** Transmits more data in limited spectrum (e.g., 4096-QAM encodes 12 bits/symbol).
- **Scalability:** Supports evolving standards like 800G/1.6T coherent optics.
- **Noise Resilience:** Coherent receivers mitigate fibre nonlinearities and signal degradation.

3.2.4 Challenges

- **Sensitivity to Noise:** Higher-order QAM (e.g., 256-QAM) requires a high signal-to-noise ratio (SNR).
- **Complex Hardware:** Linear amplifiers and precise DSP are needed to maintain signal integrity.
- **Power Consumption:** Higher-order QAM increases energy use, impacting cost and heat management.

3.2.5 Example Applications

1. **Long-Haul Fiber Networks:**
 - 16-QAM and 64-QAM for balancing speed and reach.
2. **Data Centres:**
 - 400G/800G coherent optics with 32-QAM or 64-QAM.
3. **5G Backhaul:**
 - QAM-enabled fibre links support ultra-low-latency mobile networks

3.3 Optical to Signal Noise Ratio (OSNR)

OSNR is a critical metric in fibre optic communications that quantifies the quality of an optical signal by comparing its power to the noise power within a defined bandwidth. It is foundational for ensuring reliable data transmission, especially in high-capacity systems like DWDM or OTN in fibre optical communication.

OSNR measures how much the desired optical signal stands out against background noise. It is calculated as:

$$\text{OSNR (dB)} = 10 \log_{10} \left(\frac{\text{Signal Power (S)}}{\text{Noise Power (N)}} \right)$$

- **Signal Power:** The strength of the transmitted optical signal.
- **Noise Power:** Unwanted interference from amplified spontaneous emission (ASE) in components like erbium-doped fibre amplifiers (EDFAs).
- A higher OSNR (>15–18 dB at the receiver) indicates clearer signals and better system performance.

3.3.1 Why is OSNR Important?

- **Error-Free Transmission:** Low OSNR leads to high **Bit Error Rate (BER)**, causing data corruption.
- **Long-Haul Systems:** In DWDM, OSNR determines how far a signal can travel before regeneration.
- **Performance Benchmark:** Directly correlates with system health and reliability.

3.3.2 Factors effecting OSNR?

Factor	Impact on OSNR
Amplifier Noise	EDFAs boost signals but add ASE noise, reducing OSNR over cascaded spans.
Fiber Loss	Attenuation weakens signals, lowering OSNR unless compensated.
Nonlinear Effects	High power causes distortion (e.g., four-wave mixing), degrading signal quality.
Crosstalk	Interference between DWDM channels introduces noise.

3.3.2 Measuring OSNR

- **Optical Spectrum Analyzer (OSA):** Measures signal and noise power within a reference bandwidth (typically 0.1 nm or 12.5 GHz).
- **Standards:** IEC 61280-2-9 defines methods like the interpolation (out-of-band) technique.

3.3.3 Key Takeaways

- OSNR is the **signal quality lifeline** in fibre optics.
- Critical for balancing **speed, reach, and reliability** in modern networks.
- Monitoring and improving OSNR is essential for scaling optical systems.

3.4 Attenuation

Attenuation refers to the reduction in signal strength or intensity as it travels through a medium. In fibre optics, it is measured in decibels per kilometre (dB/km) and is caused by absorption or scattering of light within the fibre

Causes:

- **Absorption:** Light energy is absorbed by impurities in the fibre.
- **Scattering:** Light is scattered by minute particles or variations in the refractive index of the fibre.

Impact: Attenuation limits the distance signals can travel without amplification. High-quality fibres have lower attenuation rates, allowing for longer transmission distances.

3.4.1 Attenuation Calculations

The total attenuation in an optical fibre link is calculated as:

$$\text{Total Attenuation (dB)} = \alpha \times L + \text{Connector Loss} + \text{Splice Loss} + \text{Bending Loss} + \text{Margin}$$

Where:

- α : Fiber attenuation coefficient (dB/km).
- L : Fiber length (km).
- **Connector Loss:** Typically, **0.3 dB per connector** (e.g., SC/LC connectors).
- **Splice Loss:** **0.1–0.3 dB per splice** (fusion or mechanical).
- **Bending Loss:** **< 0.1 dB** for gentle bends, higher for sharp bends.
- **Margin:** **Safety margin (3–5 dB)** for aging and environmental factors.

3.4.2 Attenuation Coefficient (α)

The fibre attenuation coefficient depends on the wavelength and fibre type:

Wavelength	Fiber Type	Attenuation Coefficient (α)
850 nm	Multimode (MMF)	2.5 - 3.5 dB/km
1310 nm	Single-mode (SMF)	0.3 - 0.5 dB/km
1550 nm	Single-mode (SMF)	0.2 - 0.25 dB/km

3.4.3 Example Calculation

Scenario: A 50 km single-mode fibre link operating at 1550 nm has:

- 2 connectors (0.3 dB each).
 - 5 splices (0.1 dB each).
 - Bending loss of 0.2 dB.
 - Safety margin of 3 dB.
- **Step 1:** Fiber attenuation:

$$\alpha \cdot L = 0.25 \text{ dB/km} \times 50 \text{ km} = 12.5 \text{ dB}$$
 - **Step 2:** Connector loss:

$$2 \times 0.3 \text{ dB} = 0.6 \text{ dB}$$
 - **Step 3:** Splice loss:

$$5 \times 0.1 \text{ dB} = 0.5 \text{ dB}$$
 - **Step 4:** Total attenuation:

$$12.5 + 0.6 + 0.5 + 0.2 + 3 = 16.8 \text{ dB}$$

Conclusion: The total attenuation is **16.8 dB**. The transmitter must provide enough power to overcome this loss while meeting the receiver's sensitivity.

3.4.4 Power Budget Calculation

To ensure reliable communication:

$$\text{Power Budget (dB)} = P_{\text{transmitter}} - P_{\text{receiver}} \geq \text{Total Attenuation}$$

- $P_{\text{transmitter}}$: Output power (e.g., **0 dBm** for a typical laser).
- P_{receiver} : Receiver sensitivity (e.g., **-28 dBm** for a 10 Gbps system).

Example:

$$\text{Power Budget} = 0 \text{ dBm} - (-28 \text{ dBm}) = 28 \text{ dB} \geq 16.8 \text{ dB (feasible)}$$

3.4.5. Key Factors Affecting Attenuation

1. **Material Absorption:**

- Intrinsic (silica impurities) and extrinsic (OH^- ions).

2. **Scattering Loss:**

- Rayleigh scattering (dominant at shorter wavelengths).

3. **Bending Loss:**

- Macro-bends (large-radius bends) and micro-bends (fiber jacket pressure).

4. **Splice/Connector Loss:**

- Misalignment or contamination during installation.

3.5 Dispersion Management

Dispersion in optical fibres refers to the **broadening of light pulses** as they travel through the fibre, which **degrades signal quality** and limits transmission distances. Effective dispersion management is critical for high-speed, long-distance communication systems like **DWDM** (Dense Wavelength Division Multiplexing) and **OTN** (Optical Transport Network). Below is a detailed breakdown of dispersion types, effects, and compensation strategies

3.5.1 Types of Dispersion

1. **Chromatic Dispersion**

- **Cause:** Variation in the refractive index of the fibre material with wavelength (material dispersion) and waveguide structure (waveguide dispersion).

- **Impact:** Different wavelengths travel at different speeds, causing pulse broadening. Chromatic dispersion is wavelength-dependent and increases with fibre length.

2. Polarization Mode Dispersion (PMD)

- **Cause:** Imperfections in the fiber's geometry or stress, leading to different propagation speeds for orthogonal polarization modes.
- **Impact:** Limits transmission in high-speed (>10 Gbps) systems.

3. Modal Dispersion

- **Cause:** Multiple propagation paths (modes) in multimode fibres.
- **Impact:** Only relevant in multimode fibres, not single-mode fibres.

3.5.2 Effects of Dispersion

- **Pulse Broadening:** Distorts signals, leading to intersymbol interference (ISI).
- **Increased Bit Error Rate (BER):** Degrades signal integrity.
- **Bandwidth Limitation:** Reduces the number of channels in DWDM systems.

3.5.3 Dispersion Compensation Techniques

1. Dispersion Compensating Fiber (DCF)

- **Principle:** Uses fibers with negative dispersion to counteract the positive dispersion of standard single-mode fiber (SMF).
- **Formula:**

$$D_{\text{SMF}} \cdot L_{\text{SMF}} + D_{\text{DCF}} \cdot L_{\text{DCF}} = 0$$

Example: A 100 km SMF (dispersion $D=+17 \text{ ps/nm/km}$) requires 20 km DCF ($D=-85 \text{ ps/nm/km}$) for full compensation.

- **Advantages:** Simple integration with existing SMF links.
- **Drawbacks:** High insertion loss, nonlinear effects (e.g., four-wave mixing).

2. Fiber Bragg Grating (FBG)

- **Principle:** A passive device with a periodic refractive index modulation in the fiber core. Reflects specific wavelengths to compensate for dispersion.
- **Applications:**
 - **Chirped FBG:** Compensates chromatic dispersion by reflecting different wavelengths at different positions.
 - **Slope Compensation:** Addresses dispersion variations across DWDM channels.
- **Advantages:** Low cost, low loss, and compatibility with SMF.

3. Electronic Dispersion Compensation (EDC)

- **Principle:** Uses digital signal processing (DSP) algorithms to reverse dispersion effects electronically.
- **Methods:**
 - **Transversal Filters:** Adjust weights dynamically to adapt to channel characteristics.
 - **Maximum Likelihood Sequence Estimation (MLSE):** Mitigates intersymbol interference.
- **Applications:** Common in 10–100 Gbps coherent systems.

4. Dispersion-Shifted Fibers

- **Non-Zero Dispersion-Shifted Fiber (NZ-DSF):** Shifts the zero-dispersion wavelength (λ_0) away from 1550 nm to minimize nonlinear effects.
- **Dispersion-Flattened Fiber (DFF):** Reduces dispersion slope for uniform compensation across DWDM channels.

5. Hybrid Techniques

- **DCF + EDFA:** Combines dispersion compensation with amplification (Erbium-Doped Fiber Amplifier) to extend reach.
- **FBG + WDM:** Integrates F

Method	Compensation Range	Cost	Insertion Loss	Applications
DCF	High	Moderate	High	Long-haul, legacy SMF upgrades
FBG	Moderate	Low	Low	Metro, DWDM, sensors
EDC	Moderate	High	Negligible	Coherent systems, short reach
NZ-DSF/DFF	High	High	Low	Undersea cables, DWDM

3.5.4 Key Challenges

- **Nonlinear Effects:** DCF and FBG can introduce nonlinearities like cross-phase modulation (XPM) and self-phase modulation (SPM).
- **Cost vs. Performance:** FBG and EDC are cost-effective for metro networks, while DCF is preferred for long-haul systems.
- **Temperature Sensitivity:** FBG performance varies with temperature, requiring stabilization.